

CFD SIMULATION REPORT

Replication Study of Vortex Drop Shaft

Based on Zhao et al. (2006)

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ABSTRACT

This report presents a Computational Fluid Dynamics (CFD) simulation of a vortex drop shaft based on the experimental study conducted by Zhao et al. (2006). The objective was to investigate the hydraulic behaviour of the vortex drop shaft and evaluate the capability of CFD to reproduce the experimental flow characteristics. The three-dimensional geometry was developed in **ANSYS SpaceClaim**, meshed using **ANSYS Meshing**, and analysed in **ANSYS Fluent 2026 R1** using the **Volume of Fluid (VOF)** multiphase model and the **Realizable $k-\epsilon$** turbulence model.

The simulation was used to examine the pressure distribution, velocity field, wall shear stress, turbulent dissipation rate, and energy dissipation within the shaft. The numerical results successfully reproduced the characteristic swirling flow and central air core observed in the experimental study. A qualitative comparison with the published experimental results showed good agreement in the overall flow behaviour, although differences were observed in the predicted pressure distribution and energy dissipation efficiency. These discrepancies were attributed to numerical modelling assumptions, mesh limitations, transient flow characteristics, and software constraints.

Overall, the study demonstrates that CFD provides an effective tool for analysing the complex hydraulic behaviour of vortex drop shafts and offers valuable insight into flow patterns and energy dissipation mechanisms.

Keywords: Computational Fluid Dynamics (CFD), Vortex Drop Shaft, ANSYS Fluent, Volume of Fluid (VOF), Energy Dissipation, Multiphase Flow.

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CHAPTER 1

Introduction and Objectives

1.1 Introduction

Vortex drop shafts are hydraulic structures used to convey water from a higher elevation to a lower elevation while dissipating excess energy safely. Unlike conventional plunge-flow shafts, vortex drop shafts introduce the incoming flow tangentially into a circular chamber, creating a swirling motion that forces the water to spiral along the shaft wall while maintaining a central air core. This flow pattern improves hydraulic stability, reduces cavitation risk, and provides efficient energy dissipation.

Vortex drop shafts are commonly employed in **urban stormwater drainage systems, combined sewer networks, hydropower projects, flood control structures, and deep tunnel conveyance systems** where water must be transferred safely across significant elevation differences while minimizing structural damage caused by high flow velocities.

Experimental studies provide valuable information about the hydraulic behaviour of vortex drop shafts; however, they are often expensive and time-consuming. Computational Fluid Dynamics (CFD) offers an efficient alternative by allowing detailed visualization of pressure, velocity, turbulence, and multiphase flow inside the structure while reducing the need for extensive physical testing.

The present study reproduces the experimental work of Zhao et al. (2006) using ANSYS Fluent 2026 R1. A three-dimensional model of the vortex drop shaft was developed in ANSYS SpaceClaim, meshed using ANSYS Meshing, and analysed using a transient Volume of Fluid (VOF) multiphase model with the Realizable $k-\epsilon$ turbulence model.

The numerical results obtained from the CFD simulation are compared with the experimental observations reported by Zhao et al. to evaluate the ability of the numerical model to reproduce the hydraulic characteristics of vortex flow inside the shaft.

1.2 Objective and Scope

The objectives of this study are:

- Develop the vortex drop shaft geometry in ANSYS SpaceClaim.
- Generate a suitable computational mesh for CFD analysis.
- Simulate air–water flow using the Volume of Fluid (VOF) model.
- Analyse pressure, velocity, wall shear stress, and turbulent dissipation.
- Compare CFD results with the experimental observations of Zhao et al. (2006).

The scope of this project includes the numerical simulation of a single operating condition corresponding to the design discharge reported in Zhao et al. (2006). The analysis focuses on

the flow behaviour inside the vortex drop shaft, including pressure distribution, velocity field, wall shear stress, turbulence characteristics, and air–water interaction. The CFD results are qualitatively compared with the published experimental data to assess the accuracy of the numerical model.

Chapter 2

Literature Review - Summary of Zhao et al. (2006)

The present CFD study is based on the experimental investigation conducted by **Zhao et al. (2006)**, published in the *Journal of Hydraulic Engineering*. The study examined the hydraulic behaviour of a vortex drop shaft designed to dissipate energy through a swirling flow generated by a tangential inlet.

The experimental model consisted of a rectangular approach channel connected to a volute chamber, which directed the flow into a vertical shaft through a tapered transition. As the water entered tangentially, a stable vortex developed along the shaft wall while maintaining an air core at the centre. The structure also included an outlet tunnel, an aerator, and five vent pipes to release entrained air.

Measurements were carried out for wall pressure, flow thickness, air entrainment, and energy dissipation under different flow conditions. These results serve as the benchmark for validating the CFD model developed in the present study.

2.1 Experimental Model

The important geometric and hydraulic parameters reported by Zhao et al. (2006) are summarized in Table 2.1.

Table 2.1 Experimental and Design Parameters (Zhao et al., 2006)

Parameter	Value
Shaft Diameter (D)	0.223 m
Shaft Length	2.70 m
Total Drop Height	3.15 m
Volute Chamber Diameter	1.25D
Transition Cone Slope	10:1
Design Discharge	0.051 m ³ /s
Inlet Velocity	3.92 m/s
Inlet Water Depth	0.130 m
Froude Number	3.5
Number of Vent Pipes	5

Overall Energy Dissipation	~90 %
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2.2 Major Experimental Observations

The major findings reported in the experimental study are summarized below.

Table 2.2 Key Experimental Findings

Parameter	Observation
Flow Pattern	Stable annular vortex with central air core
Wall Pressure	Positive throughout the shaft
Flow Thickness	Decreases near inlet and becomes nearly constant downstream
Hydraulic Jump	Formed approximately 0.5 m above the shaft base
Energy Dissipation	Approximately 90% overall
Air Entrainment	Increased with flow velocity

2.3 Relevance to the Present Study

The experimental study by Zhao et al. provides a reliable reference for validating CFD simulations of vortex drop shafts. In the present work, the same geometric dimensions, inlet velocity, and operating conditions were adopted to reproduce the experimental flow behaviour using ANSYS Fluent.

The CFD results presented in later chapters are compared with the experimental trends in terms of:

- Velocity distribution
- Pressure distribution
- Wall shear stress
- Turbulent dissipation
- Overall flow behaviour

The comparison helps evaluate the accuracy of the numerical model and identify possible sources of deviation between the simulation and the experimental observations.

Chapter 3

CFD Modelling Methodology

This chapter describes the methodology adopted to develop the Computational Fluid Dynamics (CFD) model of the vortex drop shaft. The modelling procedure included geometry creation, mesh generation, specification of material properties and boundary conditions, selection of numerical models, and transient simulation using ANSYS Fluent 2026 R1. The simulation setup was based on the experimental study of Zhao et al. (2006).

3.1 Geometry Development

The computational geometry was developed in **ANSYS SpaceClaim** by replicating the experimental vortex drop shaft reported by Zhao et al. (2006). The model was designed to reproduce the hydraulic behaviour of the experimental setup while maintaining the original geometric proportions and flow path.

Water enters the structure through a **rectangular approach channel**, carrying a supercritical flow with a Froude number of approximately **3.5**. The channel connects tangentially to a **circular volute chamber**, where the incoming flow is redirected by a quarter-ellipse guide wall and a deflector. This arrangement imparts angular momentum to the flow and initiates the formation of the vortex before it enters the vertical shaft.

The flow then passes through a **10:1 tapered transition cone**, which gradually reduces the chamber diameter to the shaft diameter. This smooth contraction minimizes flow separation and promotes the development of a stable annular flow inside the shaft. The **vertical shaft**, having a diameter of **0.223 m** and a length of **2.70 m**, forms the primary flow passage where the water spirals downward while maintaining a central air core.

At the base of the shaft, the flow enters a **rectangular outlet tunnel with an arched ceiling and a sump**, allowing the swirling flow to transition into horizontal discharge. An **aerator** was provided above the volute chamber to maintain atmospheric pressure within the vortex core, while **five vent pipes** located along the outlet tunnel allow entrained air to escape from the system, matching the configuration used in the experimental study.

The main dimensions of the computational model are summarized in Table 3.1.

Table 3.1 Geometry Specifications

Parameter	Value
Shaft Diameter	0.223 m
Shaft Length	2.70 m
Total Drop Height	3.15 m

Volute Chamber Diameter	1.25D (0.279 m)
Transition Cone	10:1
Approach Flow	Supercritical ($Fr = 3.5$)
Vent Pipe Diameter	40 mm
Number of Vent Pipes	5
Vent Pipe Locations	1.25De, 3De, 7De, 11.25De, 15De

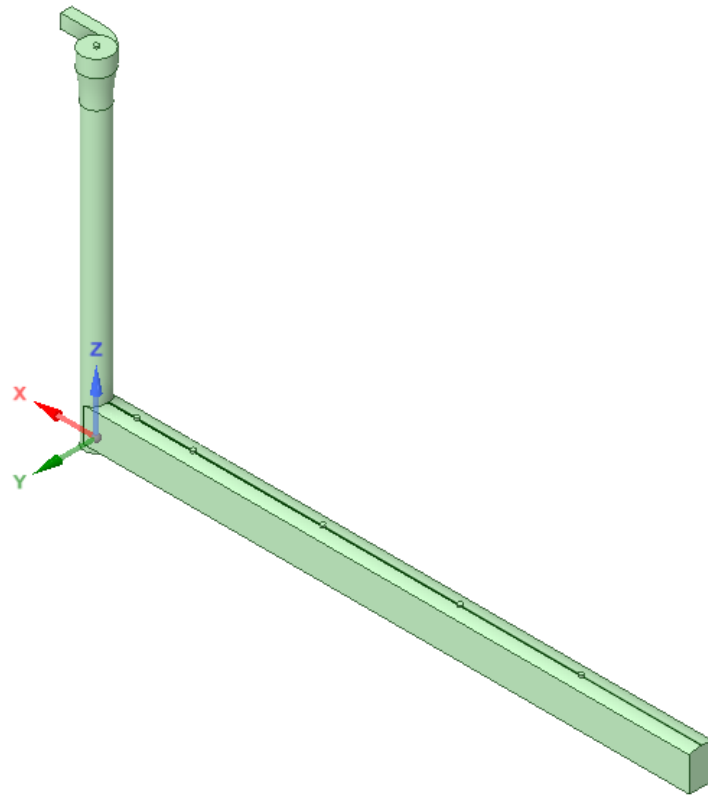


Figure 3.1 Three-dimensional geometry developed in ANSYS SpaceClaim.

3.2 Mesh Generation

The computational domain was discretized using **ANSYS Meshing** with an unstructured, predominantly tetrahedral mesh. A global element size of **0.015 m** was adopted to achieve a balance between computational cost and solution accuracy. The **Patch Conforming** meshing method was used to accurately capture the curved geometry of the volute chamber, transition cone, vertical shaft, and outlet tunnel. Automatic local refinement was generated in regions with high geometric curvature to preserve the geometry while maintaining a manageable cell count.

The final mesh statistics are presented in Table 3.2.

Table 3.2 Mesh Statistics

Parameter	Value
Mesh Method	Patch Conforming
Element Type	Mixed Cell (Predominantly Tetrahedral)
Global Element Size	0.015 m
Total Cells	595,862
Total Faces	1,225,562
Total Nodes	113,781

The mesh quality was evaluated using orthogonal quality and aspect ratio, as summarized in Table 3.3.

Table 3.3 Mesh Quality

Parameter	Value
Cell Zone	fff_solid
Zone Type	Mixed Cell
Minimum Orthogonal Quality	0.0927
Maximum Aspect Ratio	33.97

Although a small number of cells near the volute-shaft transition exhibited relatively low orthogonal quality, the majority of the computational domain maintained acceptable mesh quality for transient multiphase flow analysis

Mesh Quality

Name	Type	Min Orthogonal Quality	Max Aspect Ratio
fff_solid	Mixed Cell	0.092668553	33.971232

Orthogonal Quality

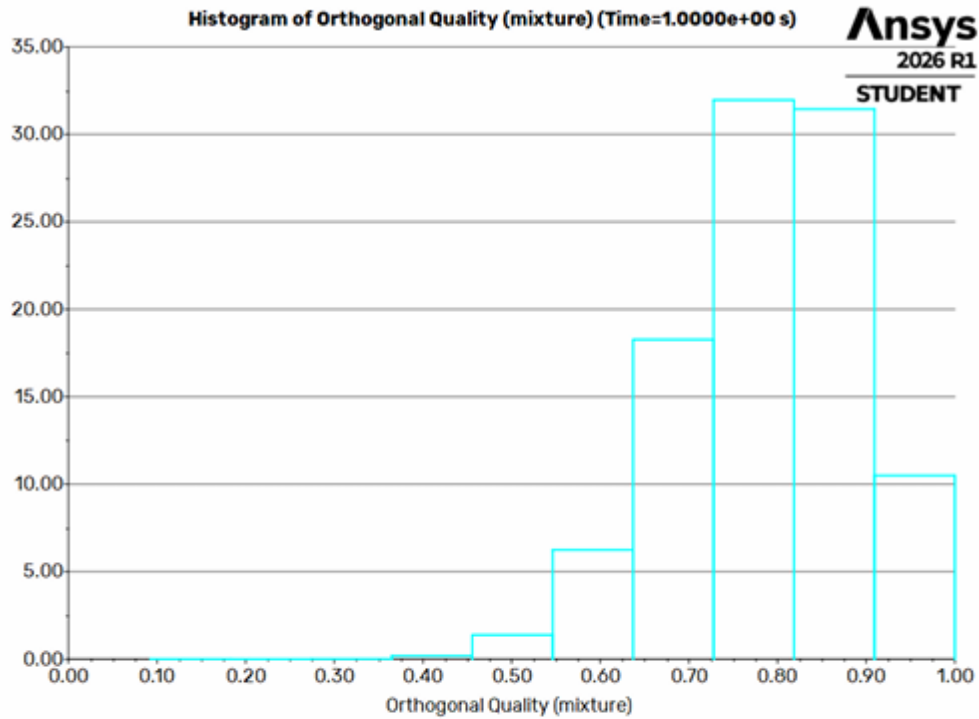


Figure 3.3. Orthogonal quality histogram of the computational mesh.

3.3 Physics Models

The simulation was performed in ANSYS Fluent 2026 R1 using a pressure-based, transient solver. Since the problem involves simultaneous air-water flow with a moving free surface, the Volume of Fluid (VOF) model was adopted. Turbulence was modelled using the Realizable $k-\epsilon$ model with standard wall functions to capture the highly turbulent swirling flow within the shaft.

Table 3.4 Physics Models

Model	Setting
Solver	Pressure-Based
Space	3D
Time	Transient (Unsteady)
Time Formulation	First-Order Implicit
Precision	Double Precision
Multiphase Model	Volume of Fluid (VOF)
Interface Modelling	Sharp Interface with Interfacial Anti-Diffusion
VOF Scheme	Compressive
Turbulence Model	Realizable k- ϵ
Wall Treatment	Standard Wall Functions
Gravity	-9.81 m/s ² (Z-direction)

3.4 Material Properties

The computational domain consisted of two immiscible fluid phases: air and water. The material properties were obtained from the ANSYS Fluent material database and are listed in Table 3.5.

Table 3.5 Material Properties

Material	Density (kg/m ³)	Dynamic Viscosity (Pa·s)	Phase
Air	1.225	1.7894×10^{-5}	Primary
Water	998.2	1.003×10^{-3}	Secondary

3.5 Boundary Conditions

Boundary conditions were assigned to closely replicate the experimental setup described by Zhao et al. (2006). A velocity inlet was prescribed at the approach channel, while atmospheric pressure conditions were specified at the aerator, outlet tunnel, and vent pipes. All solid boundaries were modelled as stationary no-slip walls.

Table 3.6 Boundary Conditions

Boundary	Type	Specification
Inlet	Velocity Inlet	Velocity = 3.92 m/s , Turbulence Intensity = 5% , Turbulent Viscosity Ratio = 10
Aerator	Pressure Inlet	Gauge Pressure = 0 Pa
Outlet Tunnel	Pressure Outlet	Gauge Pressure = 0 Pa
Vent Pipes (1–5)	Pressure Outlet	Gauge Pressure = 0 Pa
All Walls	Stationary Wall	No-slip, Roughness Height = 0 m

3.6 Solver Settings

The numerical solution employed the **PISO** pressure-velocity coupling algorithm with second-order spatial discretization schemes for momentum and **PRESTO!** pressure interpolation. The principal solver settings are summarized in Tables 3.7-3.9.

Table 3.7 Solution Methods

Parameter	Setting
Pressure–Velocity Coupling	PISO
Pressure Discretization	PRESTO!
Momentum	Second Order Upwind
Volume Fraction	Compressive
Turbulent Kinetic Energy (k)	First Order Upwind
Turbulent Dissipation Rate (ϵ)	First Order Upwind
Absolute Velocity Formulation	True

Table 3.8 Under-Relaxation Factors

Parameter	Value(Initial Run)	Value(Updated Run)
Pressure	0.3	0.3
Density	1.0	1.0
Body Forces	1.0	1.0
Momentum	0.5	0.7
Volume Fraction	0.5	0.5
Turbulent Kinetic Energy	0.5	0.8
Turbulent Dissipation Rate	0.5	0.8
Turbulent Viscosity	0.8	1.0

Table 3.9 Run Calculation Parameters

Parameter	Initial Run	Updated Run
Number of Time Steps	1000	2500
Time Step Size	0.001 s	0.001 s
Max Iterations per Time Step	20	20
Total Simulation Flow Time	1.0 s	2.5 s
Data Sampling for Time Statistics	No	Yes
Autosave Frequency	Every 50 steps	Every 50 steps
Initialization	Hybrid	Hybrid + Patch (Water VF=0 in shaft)

3.7 Post-Processing

The simulation results were analysed using the post-processing tools available in **ANSYS Fluent**. Contour plots, vector plots, and XY plots were generated to evaluate the hydraulic performance of the vortex drop shaft and compare the numerical predictions with the experimental observations.

Table 3.10 Post-Processing Outputs

Output	Purpose
Mesh Visualization	Verification of computational mesh
Static Pressure Contours	Pressure distribution within the shaft
Velocity Magnitude Contours	Flow acceleration and vortex formation
Wall Shear Stress Contours	Identification of high shear regions
Turbulent Dissipation Rate	Assessment of energy dissipation
Velocity Vector Plots	Visualization of flow direction
Velocity XY Plot	Velocity variation along the shaft
Pressure XY Plot	Pressure variation along the shaft

Table 3.1 Solution Convergence Status

Residual	Final Value	Criterion	Status
Continuity	0.001150	0.001	Not Converged
X-Velocity	5.44×10^{-7}	0.001	Converged
Y-Velocity	5.17×10^{-7}	0.001	Converged
Z-Velocity	5.31×10^{-7}	0.001	Converged
Turbulent Kinetic Energy (k)	1.40×10^{-6}	0.001	Converged
Turbulent Dissipation Rate (ϵ)	1.72×10^{-6}	0.001	Converged
Volume Fraction (Water VF)	0	0.001	Converged

Note: The continuity residual (0.00115) slightly exceeds the convergence criterion of 0.001 at the end of the run. This is expected behaviour for VOF multiphase transient simulations with a free surface, particularly with the relatively coarse mesh and short simulation time. All other residuals are well converged. The continuity residual would be expected to decrease further with additional time steps.

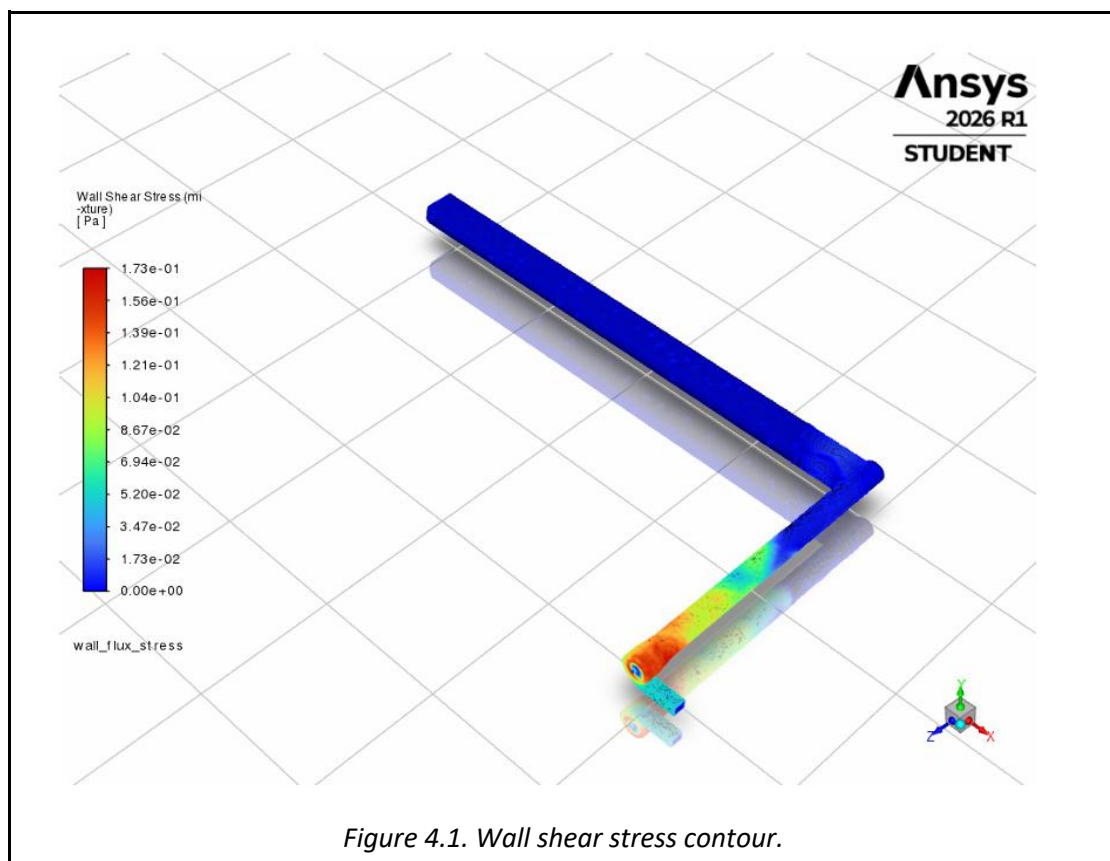
Chapter 4

Results and Discussion

4.1 Wall Shear Stress

Wall shear stress represents the tangential force exerted by the fluid on the solid boundaries and provides an indication of regions experiencing high frictional losses. Figure 4.1 shows that wall shear stress is relatively low along the upper shaft but increases significantly towards the lower shaft and outlet tunnel. The maximum wall shear stress obtained from the simulation is 0.173 Pa.

The increase in wall shear stress near the shaft base is caused by the interaction of the high-momentum swirling flow with the wall as it changes direction before entering the outlet tunnel. The combination of flow acceleration, curvature, and increased turbulence in this region produces larger velocity gradients adjacent to the wall, resulting in higher shear stresses. These regions are also expected to contribute significantly to hydraulic energy dissipation within the structure.



4.2 Static Pressure Distribution

4.2.1 Wall Surface

The wall pressure contour illustrates the variation of static pressure along the boundaries of the vortex drop shaft. The pressure remains relatively low over the upper shaft and gradually increases towards the lower section, reaching a maximum value of approximately **37.8 Pa** near the shaft–outlet transition. Localized negative pressures of approximately **−11.4 Pa** are observed near the outlet due to flow acceleration and local recirculation.

The increase in pressure near the base results from the reduction in flow velocity as the swirling water enters the outlet tunnel. The change in flow direction and the formation of the water cushion increase the local static pressure, whereas the outlet experiences lower pressures because of the accelerated discharge of the fluid.

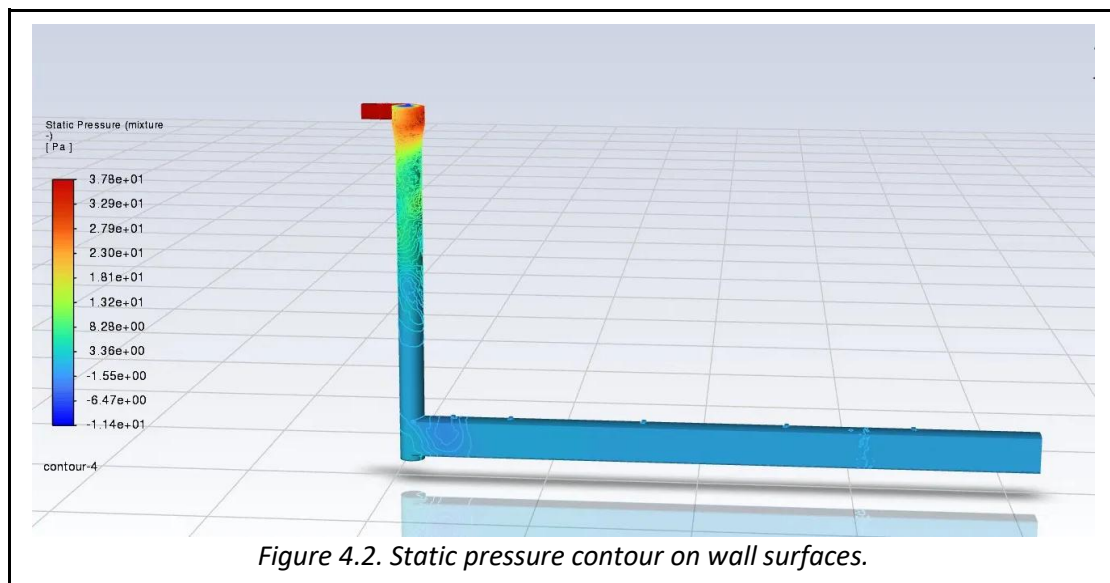
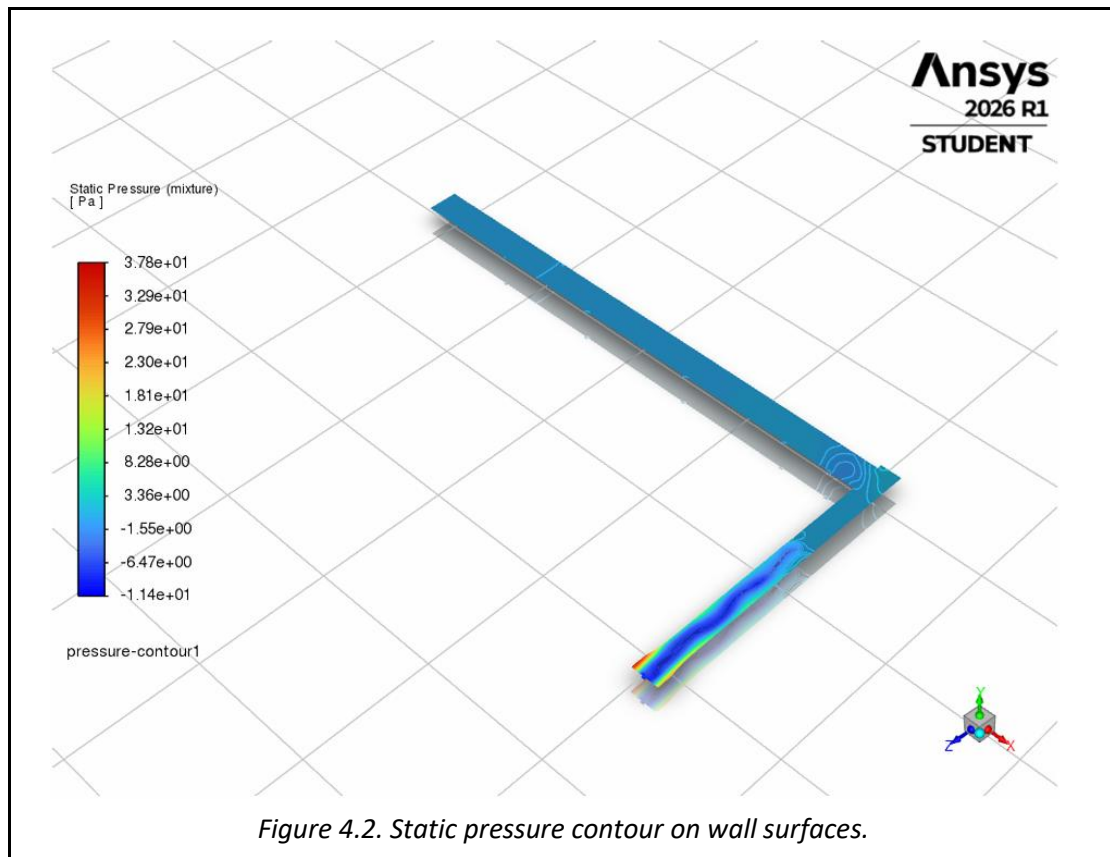


Figure 4.2. Static pressure contour on wall surfaces.

4.2.2 Mid-Plane Cross Section

The mid-plane pressure contour provides a clearer representation of the pressure variation within the fluid domain. Unlike the wall contour, the cross-sectional view reveals the internal pressure field throughout the shaft and outlet tunnel. The pressure gradually increases towards the lower shaft, while lower pressure regions remain concentrated around the vortex core where the rotating flow produces centrifugal effects.

The pressure distribution confirms the development of the vortex flow and indicates that the pressure recovery primarily occurs near the shaft base before the flow enters the outlet tunnel.

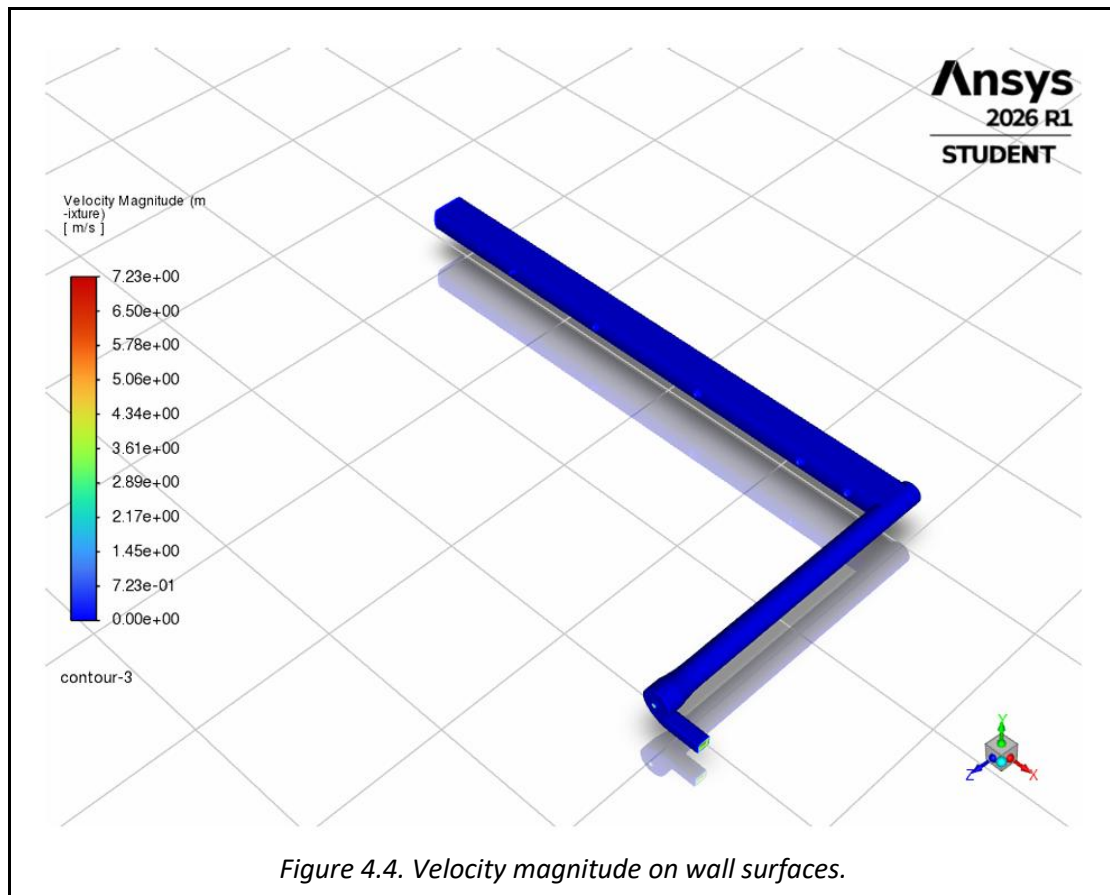


4.3 Velocity Magnitude

4.3.1 Wall Surface

The velocity contour on the wall surfaces reflects the influence of the no-slip boundary condition imposed on all solid walls. Consequently, velocities remain close to zero along most wall surfaces, except near the inlet where the incoming flow first enters the volute chamber. The maximum velocity obtained from the simulation is approximately **7.23 m/s**.

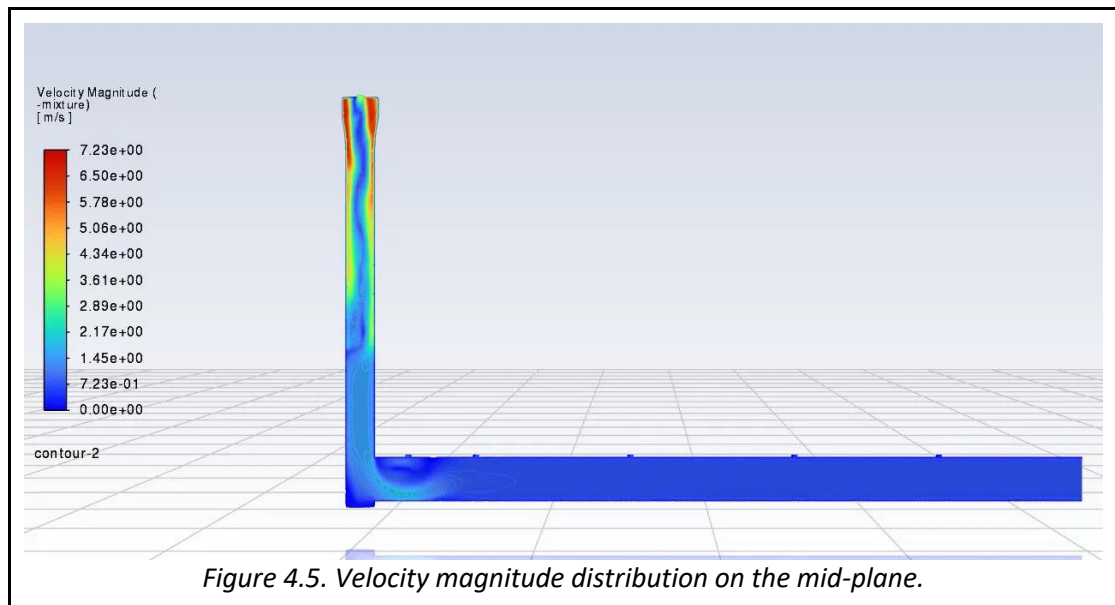
The high inlet velocity results from the supercritical flow entering the chamber, while the low wall velocities indicate that the boundary condition has been correctly enforced throughout the simulation.



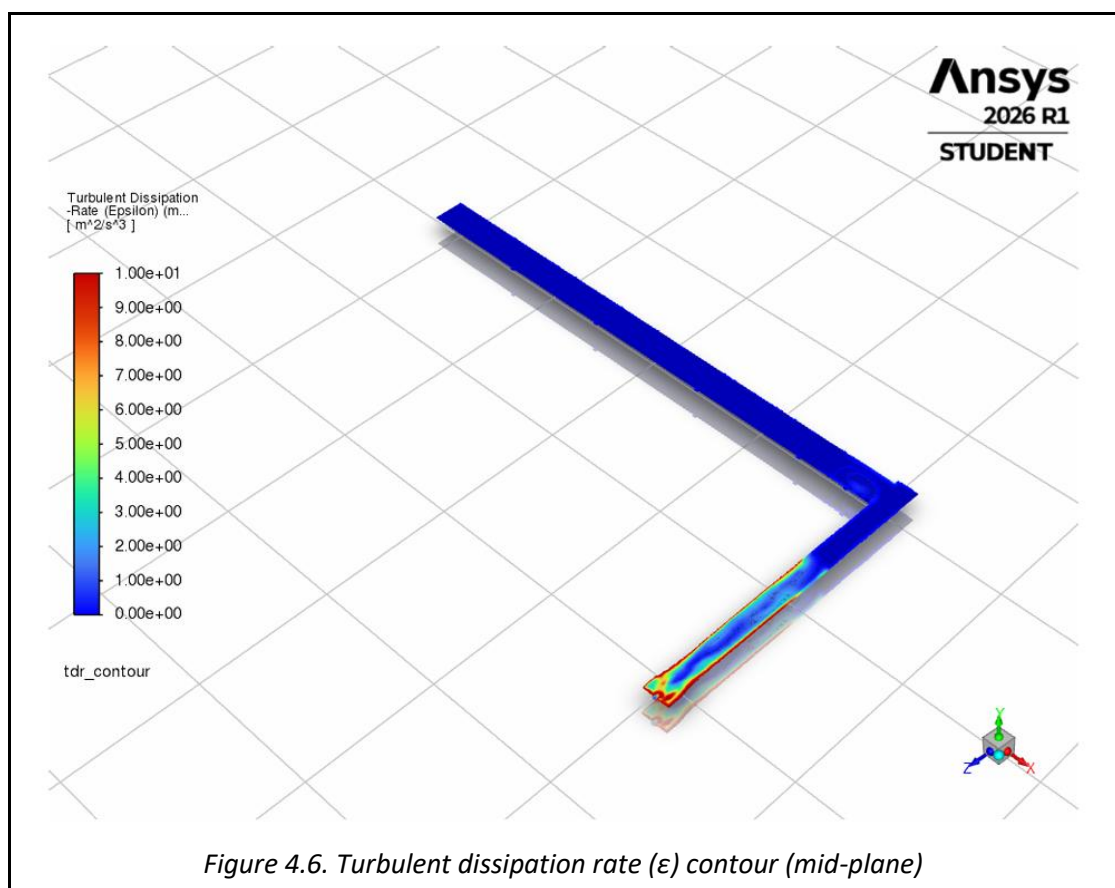
4.3.2 Mid-Plane Cross Section

The velocity contour clearly illustrates the development of the vortex inside the shaft. As the flow enters tangentially through the volute chamber, it is redirected into the vertical shaft, producing a spiralling motion around the central air core. Higher velocities are concentrated near the shaft wall where the annular water film develops, while comparatively lower velocities occur near the centre of the shaft due to the presence of the air core.

The highest velocities are observed near the volute chamber and upper shaft, where the flow first acquires its angular momentum before gradually redistributing along the shaft.



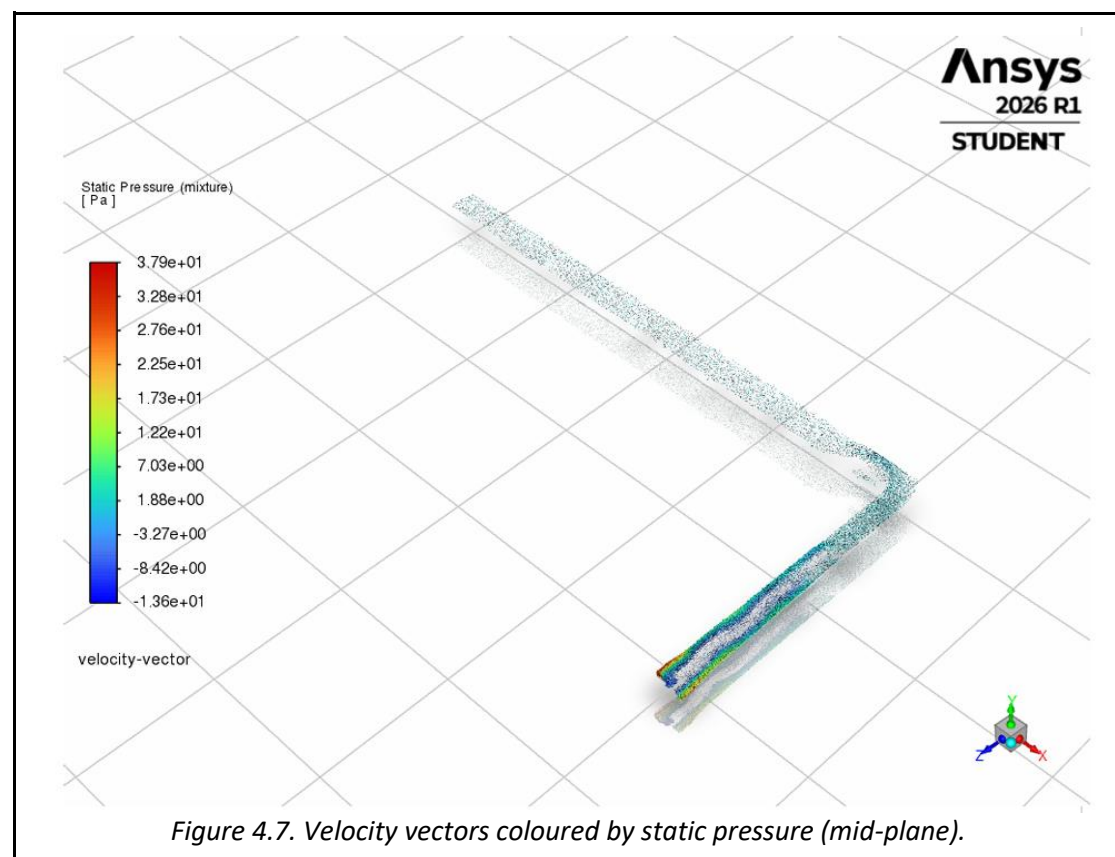
4.4 Turbulent Dissipation Rate



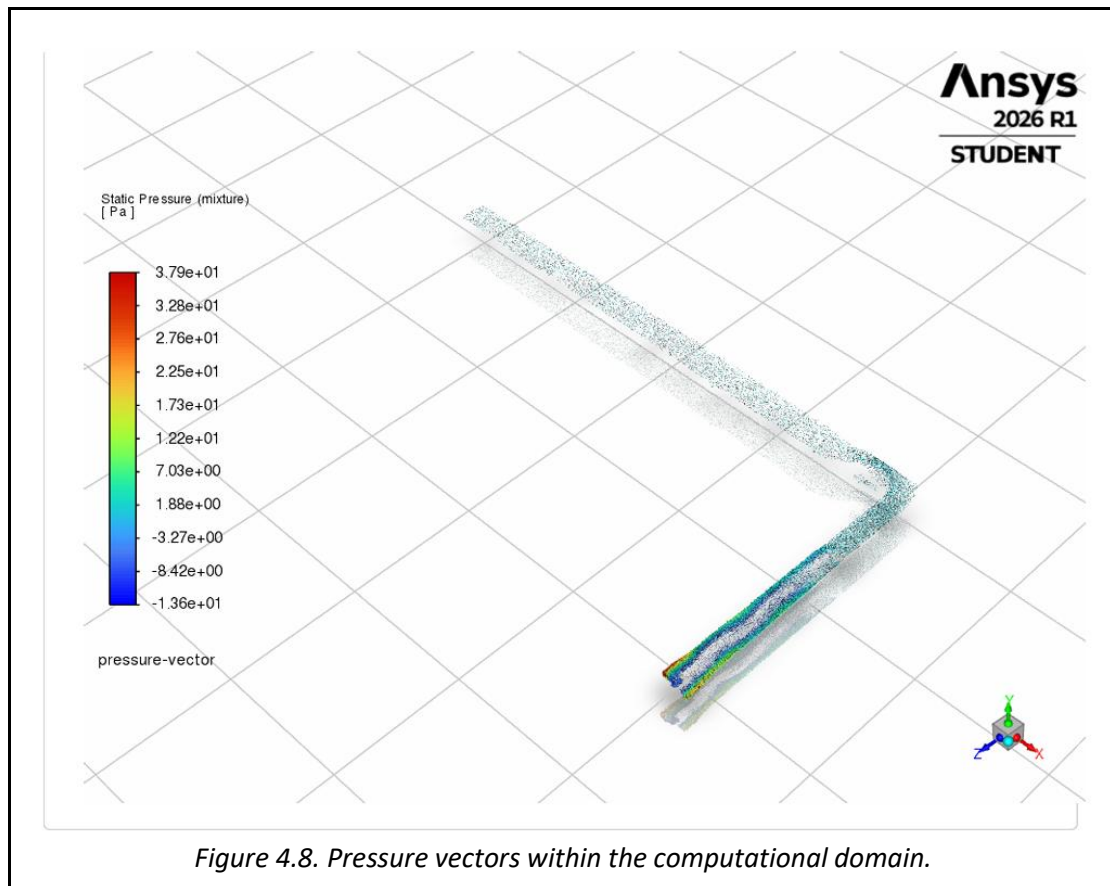
The turbulent dissipation rate represents the rate at which turbulent kinetic energy is converted into thermal energy and serves as an indicator of energy loss within the flow. Figure 4.6 shows elevated dissipation near the volute chamber, along the shaft wall, and close to the outlet tunnel.

The high dissipation observed near the volute chamber corresponds to the rapid transition of the incoming flow into a vortex. Additional dissipation occurs near the shaft wall because of the strong shear generated by the annular flow. A secondary peak near the shaft base indicates intense mixing as the swirling flow enters the outlet tunnel.

4.5 Velocity and Pressure Vectors



The velocity vector plot illustrates the direction and magnitude of the flow within the computational domain. The vectors clearly show the transition from horizontal flow in the approach channel to a vertical spiralling motion inside the shaft. The distribution confirms the successful generation of vortex flow by the tangential inlet and demonstrates the concentration of flow near the shaft wall.

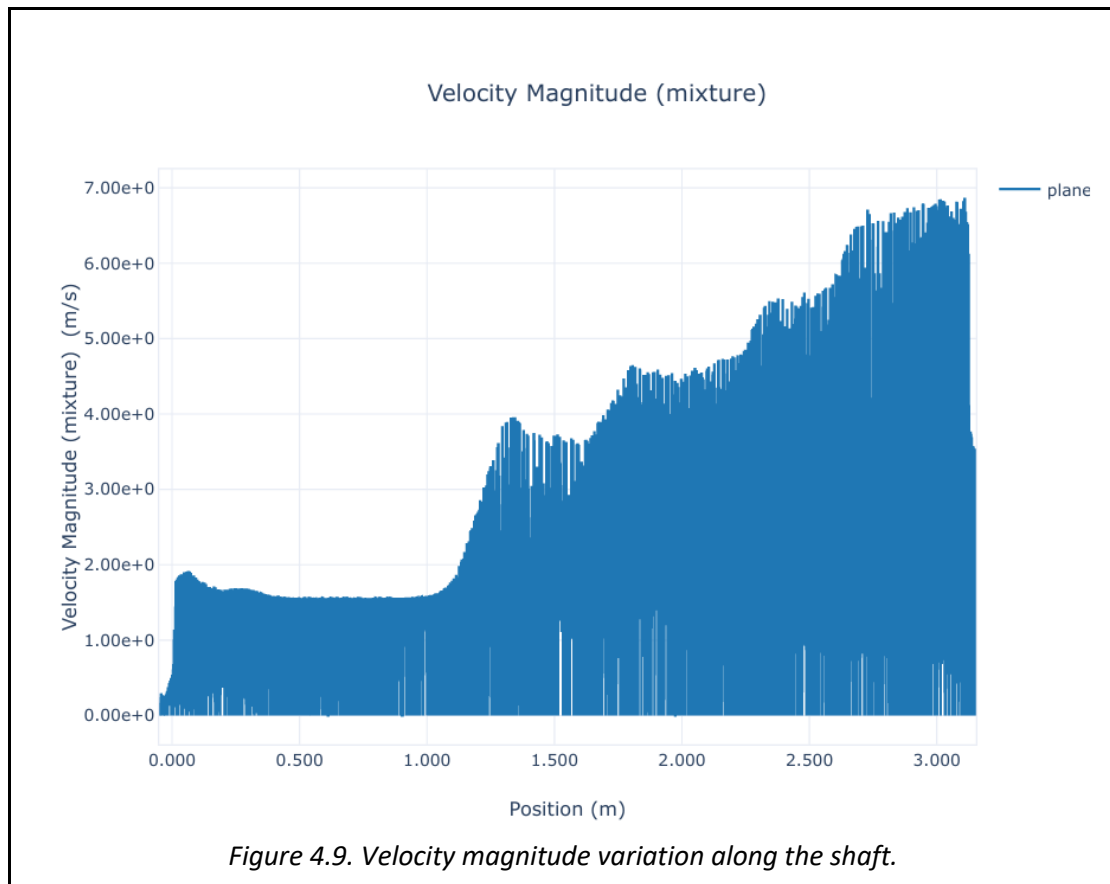


The pressure vectors indicate the pressure-driven movement of the fluid through the shaft and outlet tunnel. Higher pressure regions are concentrated near the shaft base, whereas lower pressures are associated with the central vortex core. Together, the velocity and pressure vectors provide a clear visualization of the hydraulic behaviour within the drop shaft.

4.6 Velocity Profile Along the Shaft

The velocity profile illustrates the variation of flow velocity along the vertical shaft. The velocity is highest near the inlet and gradually decreases towards the outlet as the flow loses energy through wall friction, turbulence, and mixing. The reduction in velocity reflects the energy dissipation mechanism of the vortex drop shaft, where the swirling flow progressively converts kinetic energy into turbulent energy.

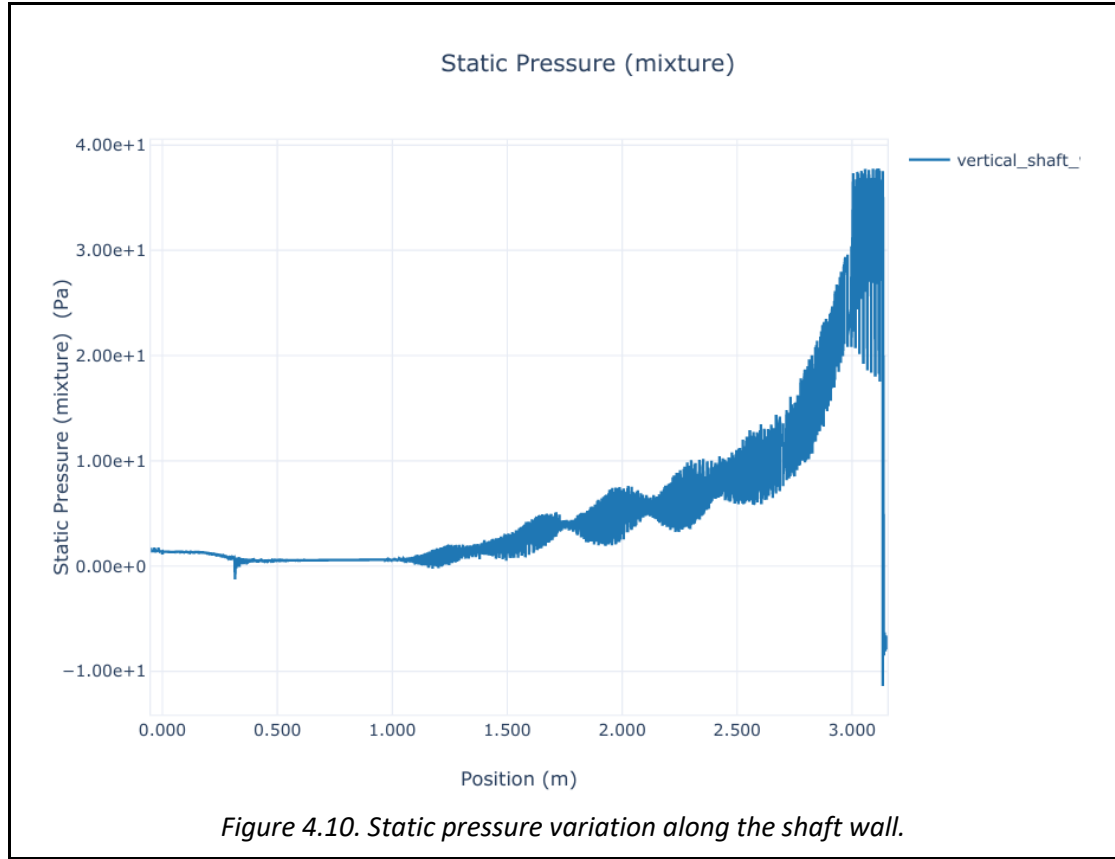
The velocity profile exhibits noticeable oscillations along the shaft rather than a smooth decrease. These fluctuations arise from the transient nature of the air-water interface captured using the Volume of Fluid (VOF) model. As the vortex rotates, the interface continuously changes position, causing local variations in velocity at the sampling line. Additional oscillations are produced by turbulent eddies, secondary circulation within the vortex, and the periodic interaction between the rotating water film and the central air core. Such fluctuations are characteristic of transient multiphase vortex flows and indicate that the flow is inherently unsteady rather than steady.



4.7 Pressure Profile Along the Shaft

The pressure profile shows an overall increase in static pressure towards the lower portion of the shaft, reflecting the combined influence of gravity, centrifugal forces, and flow deceleration before discharge into the outlet tunnel. Although the general trend follows the expected pressure recovery along the shaft, the pressure distribution is accompanied by significant oscillations.

The oscillatory behaviour is primarily associated with the transient movement of the air–water interface predicted by the VOF model. As the vortex develops, the thickness of the annular water film and the size of the central air core continuously fluctuate, producing local pressure variations along the shaft. In addition, turbulent fluctuations and the rotating nature of the vortex generate alternating regions of high and low pressure, resulting in the oscillatory pressure profile observed in the simulation. These pressure fluctuations are typical of transient swirling multiphase flows and are consistent with the highly dynamic behaviour of vortex drop shafts.



Note: The oscillations observed in the velocity and pressure profiles represent instantaneous values extracted from a transient simulation. Averaging the solution over multiple time steps after the flow reaches a periodic state would produce smoother profiles and a more representative mean flow behaviour.

4.8 Energy Dissipation Analysis

The primary objective of a vortex drop shaft is to dissipate hydraulic energy efficiently while maintaining a stable swirling flow. To evaluate the energy dissipation performance of the present CFD model, the total head at selected sections along the shaft was estimated using the Bernoulli equation:

$$H = \frac{P}{\rho g} + \frac{V^2}{2g} + Z$$

where H is the total head (m), P is the static pressure (Pa), V is the flow velocity (m/s), Z is the elevation above the sump (m), $\rho=998.2 \text{ kg/m}^3$, and $g=9.81 \text{ m/s}^2$.

Pressure and velocity values were extracted from the pressure vs. shaft length and velocity vs. shaft length plots at four locations corresponding to the sections defined by Zhao et al. (2006). Since the values were read directly from the CFD plots, they should be regarded as **approximate** and are intended primarily for qualitative comparison.

Table 4.1 Approximate Pressure and Velocity at Selected Sections

Section	Elevation, Z (m)	Velocity (m/s)	Pressure (Pa)
Section 1 (Volute Top)	3.150	6.7	35
Section 2	2.385	5.0	10
Section 3	0.785	1.6	1
Section 4 (Outlet/Sump)	0.000	0.1	1

The corresponding total head values calculated at each section are presented in Table 4.2.

Table 4.2 Total Head at Selected Sections

Section	Pressure Head (m)	Velocity Head (m)	Elevation Head (m)	Total Head, H (m)
1	0.0036	2.288	3.150	5.442
2	0.0010	1.274	2.385	3.660
3	0.0001	0.130	0.785	0.915
4	0.0001	0.0005	0.000	0.001

The percentage energy dissipation between consecutive sections was then determined from the reduction in total head.

Table 4.3 Estimated Energy Dissipation Efficiency

Segment	Energy Dissipation (%)
Section 1 → Section 2	32.7
Section 2 → Section 3	75.0
Section 3 → Section 4	99.9
Overall (Section 1 → Section 4)	99.8

The results indicate a continuous reduction in total head as the flow progresses through the vortex drop shaft, demonstrating the energy dissipating capability of the structure. The largest reduction in head occurs between **Sections 3 and 4**, where the flow enters the outlet tunnel and experiences significant deceleration and mixing.

The estimated overall energy dissipation efficiency obtained from the CFD simulation is **99.8%**, which is higher than the approximately **90%** reported by Zhao et al. (2006). Despite this quantitative difference, both the numerical and experimental studies indicate that the

majority of the hydraulic energy is dissipated within the lower portion of the shaft and outlet region.

4.9 Overall Comparison with Zhao et al. (2006)

The CFD results obtained in the present study were compared with the experimental observations reported by Zhao et al. (2006). Overall, the numerical model successfully reproduced the characteristic vortex flow pattern, central air core formation, and the general trends of pressure, velocity, and energy dissipation. However, quantitative differences were observed in several parameters, particularly the pressure distribution and calculated energy dissipation efficiency. A summary of the comparison is presented in Table 4.4.

Table 4.4 Overall Comparison Between Experimental and CFD Results

Parameter	Zhao et al. (2006)	Present CFD Study	Agreement
Inlet Velocity	3.92 m/s	3.92 m/s	Excellent
Froude Number	3.5	3.5	Excellent
Flow Pattern	Stable swirling annular flow	Stable swirling annular flow	Excellent
Central Air Core	Maintained throughout shaft	Maintained throughout shaft	Good
Pressure Trend	Increases towards shaft base	Similar trend observed	Good
Pressure Magnitude	Higher experimental values	Lower predicted values	Moderate
Velocity Distribution	High near inlet, decreases downstream	Similar trend observed	Good
Wall Shear Region	Highest near lower shaft and outlet	Highest near lower shaft and outlet	Good
Major Energy Dissipation Region	Lower shaft and outlet	Lower shaft and outlet	Good
Overall Energy Dissipation	≈90%	≈99.8%*	Moderate
Hydraulic Behaviour	Stable vortex flow	Stable vortex flow	Good

**Estimated from approximate pressure and velocity values extracted from CFD profile plots.*

Chapter 5

Discussion and Limitations

The CFD simulation successfully reproduced the overall hydraulic behaviour of the vortex drop shaft, including the formation of the swirling flow, the central air core, and the principal regions of energy dissipation. Although the qualitative flow behaviour agreed well with the experimental observations reported by Zhao et al. (2006), several quantitative differences were observed in the pressure distribution, velocity profile, and calculated energy dissipation. The possible reasons for these differences are discussed below.

5.1 Pressure Prediction

The CFD simulation reproduced the overall pressure trend within the shaft, with pressure increasing towards the lower section before the outlet tunnel. However, the predicted pressure magnitudes were significantly lower than those reported experimentally.

One contributing factor is that the simulation was performed for a limited physical time, during which the flow had not completely developed into a fully periodic vortex. The residual history also showed that the **continuity residual remained slightly above the prescribed convergence criterion ($0.00115 > 0.001$)**, indicating that the solution had not reached complete numerical convergence. Since wall pressure is highly sensitive to the stability of the swirling flow, incomplete convergence can influence the predicted pressure values.

5.2 Velocity Distribution

The numerical model successfully captured the transition from horizontal inlet flow to the characteristic annular vortex inside the shaft. Higher velocities were observed near the shaft wall, while lower velocities occurred near the central air core, consistent with the expected vortex structure.

The velocity profile, however, exhibited noticeable oscillations. These fluctuations are characteristic of transient **Volume of Fluid (VOF)** simulations, where the air-water interface continuously changes with time. The instantaneous movement of the interface, together with turbulent eddies and secondary circulation within the vortex, produces local velocity fluctuations along the sampling line. Consequently, the extracted profile represents an instantaneous flow field rather than a time-averaged solution.

5.3 Energy Dissipation

The estimated overall energy dissipation obtained from the CFD simulation was approximately **99.8%**, compared to approximately **90%** reported by Zhao et al. (2006). Although the CFD model identified the correct regions of major energy dissipation, the calculated efficiency was higher than the experimental value.

This difference is primarily associated with the very low outlet velocity predicted by the simulation. Since the total head includes a velocity head component, the reduced outlet velocity resulted in a larger decrease in total head and consequently a higher calculated energy dissipation efficiency. In addition, the energy calculations were performed using **approximate pressure and velocity values extracted from the XY plots**, introducing a degree of uncertainty into the calculated efficiencies.

5.4 Mesh and Numerical Model

The computational mesh was generated within the limitations of the **ANSYS Student Version**, which restricts the maximum mesh size to approximately **one million cells**. Therefore, the mesh was selected to balance computational accuracy and simulation time while remaining within the available software resources. Although the mesh captured the overall flow behaviour, further refinement near the volute chamber and shaft transition could improve the prediction of local pressure and velocity gradients.

The **Realizable $k-\epsilon$ turbulence model** was selected because of its robustness and computational efficiency for swirling flows. While it successfully reproduced the overall vortex behaviour, it may not fully capture the complex turbulent structures associated with strong swirling multiphase flow. More advanced models, such as the **Reynolds Stress Model (RSM)** or **Large Eddy Simulation (LES)**, could provide improved prediction accuracy but require substantially higher computational resources.

5.5 Overall Assessment

Despite the quantitative differences, the CFD model successfully reproduced the principal hydraulic characteristics of the vortex drop shaft. The formation of the vortex, the central air core, the regions of high wall shear stress, and the dominant energy dissipation zones were all consistent with the experimental observations. Therefore, the present model provides a reliable qualitative representation of the flow behaviour and demonstrates the capability of CFD for analysing complex vortex drop shaft hydraulics.

Chapter 6

Conclusion and Future Work

6.1 Conclusion

The present study successfully developed a three-dimensional CFD model of a vortex drop shaft based on the experimental configuration reported by Zhao et al. (2006). The geometry was modelled in **ANSYS SpaceClaim**, meshed using **ANSYS Meshing**, and analysed in **ANSYS Fluent** using the **Volume of Fluid (VOF)** multiphase model and the **Realizable $k-\epsilon$** turbulence model.

The simulation successfully reproduced the characteristic hydraulic behaviour of the vortex drop shaft, including the formation of the swirling annular flow, the central air core, and the major regions of energy dissipation. The pressure, velocity, wall shear stress, and turbulent dissipation contours showed trends that were consistent with the experimental observations, demonstrating that the numerical model captured the overall flow physics effectively.

Although the CFD model predicted lower pressure values and higher overall energy dissipation than the experimental study, the general flow behaviour showed good qualitative agreement. These differences are attributed to the transient nature of the simulation, numerical modelling assumptions, mesh resolution, and software limitations discussed in the previous chapter.

Overall, the study demonstrates that CFD is an effective tool for analysing vortex drop shaft hydraulics and can provide valuable insight into flow behaviour that is difficult to obtain through experimental measurements alone.

6.2 Future Work

Future studies may improve the prediction accuracy by:

- Performing a mesh independence study using a finer computational mesh.
- Extending the transient simulation time to obtain a fully developed periodic flow.
- Employing advanced turbulence models such as the **Reynolds Stress Model (RSM)** or **Large Eddy Simulation (LES)**.
- Extracting pressure and velocity values directly from monitoring locations for more accurate quantitative comparisons.
- Quantify the air core radius profile along the shaft height by performing iso-surface extraction of the air–water interface (Volume Fraction = 0.5) in ANSYS Fluent/CFD-Post. Alternatively, the air core radius may be determined using the continuity equation with section-averaged vertical velocity at multiple cross-sections, providing a more comprehensive validation of the CFD model against the experimental study.

References

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